Qns	Answer	Marks
3(a)	For a system in equilibrium, sum of clockwise moments about a (or any) point equals sum of anticlockwise moments about the same point	B1 B1
3(b)	Let the length of the rod be *L*. Take moment about the point of contact between the rod and the ground. sum of clockwise moments $1000\left(\frac{L}{2}\cos 30^o\right)$ OR $1000\left(\frac{L}{2}\sin 60^o\right)$ sum of anticlockwise moments $T\sin 17^o\left(L\right)$ OR $T\cos 73^o\left(L\right)$ **OR** sum of clockwise moments $1000\left(\frac{L}{2}\cos 30^o\right)+T\cos 47^o\left(L\sin 30^o\right)$ OR $1000\left(\frac{L}{2}\sin 60^o\right)+T\sin 43^o\left(L\cos 60^o\right)$ sum of anticlockwise moments $T\sin 47^o\left(L\cos 30^o\right)$ OR $T\cos 43^o\left(L\sin 60^o\right)$ T = 1500 N (or 1480 N)	C1 C1 (C1) (C1) A1